

ULUX — Product & Development Brief

1. Product Overview

Product name: ULUX 有練有差

ULUX is a private, mobile-first fitness training tracking web application designed primarily for serious strength-training users.

The core philosophy is:

Extremely simple on the surface, highly structured underneath.

Logging a workout should be fast and frictionless, especially when the user is standing in a gym and using a phone with one hand.

At the same time, ULUX should maintain structured historical data that can later support:

- Personal records (PRs)
- Training volume
- Exercise progression
- Workout frequency
- Muscle-group distribution
- Historical comparisons
- Advanced analytics
- Potential future AI-powered insights

The product should feel clean, calm, modern, and lightweight rather than like an overly complicated fitness dashboard.

2. Product Priorities

The priorities are:

1. **Strength training tracking**
2. **Fast workout logging**
3. **Historical training data**
4. **Progress analytics**
5. **Workout repetition / continuity**
6. General cardio and sports tracking as secondary features

ULUX is NOT intended to be a social fitness network.

There are currently:

- No followers
- No public profiles

- No social feed
- No likes/comments
- No community features

All user data is private.

3. Core UX Principle

Frontend simplicity, backend richness

The Home screen should NOT overwhelm the user with statistics.

When the user opens ULUX, the primary question should be:

"What are you training today?"

The user should be able to start logging a workout immediately.

Advanced information such as:

- PRs
- Training volume
- Progress charts
- Muscle distribution
- Historical comparisons

should live behind dedicated Progress / Analytics interfaces.

Do not put every statistic on the Home screen.

4. Authentication & Account System

ULUX uses a deliberately minimal account system.

Account creation

A new user provides:

- ID
- 4-digit PIN
- Email
- Nickname

ID

- The ID is the permanent account identifier.

- The ID must be unique.
- The ID can contain an unrestricted number of characters.
- The ID cannot be changed after account creation.
- The ID is used only for authentication/account identification.
- The system should normally address the user by their nickname, NOT their ID.

PIN

- Exactly 4 numeric digits.
- Used together with the ID for normal login.
- The PIN should never be displayed in plaintext after creation.
- PIN should be securely hashed/stored.

Email

The email is NOT required during normal login.

Normal login requires only:

ID + PIN

The email is used for account recovery.

Account recovery

If the user forgets their ID or PIN, they can use the email associated with the account to initiate recovery.

The recovery flow should allow the user to verify ownership through their email and reset their PIN.

Because the ID is immutable, the recovery interface should also reveal or recover the user's ID after successful email verification.

Do not expose account information to unauthenticated users without appropriate verification.

5. Account Settings

The first version should include a simple Settings page.

Settings should include:

- Nickname
- Change PIN
- Export data to Excel
- Manage custom exercises
- Logout

The permanent ID should be displayed as account information but should NOT have an edit function.

Email management does not need to be a primary feature in the first version unless required by the authentication/recovery implementation.

6. Units

ULUX uses **kilograms (kg) only**.

Do not implement kg/lb switching in the first version.

7. Workout Session Concept

ULUX must use the concept of a **Workout Session**.

A Workout Session represents one distinct training session.

For example:

Workout Session — September 7

- Bench Press
- Incline Bench Press
- Cable Fly
- Triceps Pushdown

A user may have multiple workout sessions on the same calendar day.

For example:

September 7

Morning

- Weight training

Evening

- Tennis

These should be stored as two separate Workout Sessions.

8. Workout Logging Flow

The primary workout creation flow should be hierarchical and contextual.

The user selects:

Step 1 — Training Type

Dropdown:

- Weight Training
 - Cardio
 - Ball Sports
 - Other Exercise
-

Step 2 — Category

The available categories depend on Step 1.

For Weight Training, categories should be organized primarily around muscle groups, such as:

- Chest
- Back
- Shoulders
- Glutes & Legs
- Arms
- Core
- Other

The exact taxonomy can be expanded as appropriate.

Important:

An exercise should NOT be restricted to one muscle group.

For example:

Face Pull

may belong to both:

- Shoulders
- Back

The underlying exercise model should support many-to-many relationships between exercises and muscle groups.

The UI may therefore surface the same exercise under multiple relevant categories.

9. Exercise Selection

After selecting a category, the user selects an exercise.

Examples:

Chest

- Flat Bench Press
- Incline Bench Press
- Push-up
- Cable Fly
- etc.

Back

- Lat Pulldown
- Pull-up
- Seated Cable Row
- Face Pull
- etc.

Glutes & Legs

- Squat
- Deadlift
- Romanian Deadlift
- Leg Press
- Hip Thrust
- etc.

Shoulders

- Overhead Press
- Lateral Raise
- Face Pull
- etc.

Exercise names should be displayed in:

Traditional Chinese + English

Example:

平胸臥推 — Flat Bench Press

Use Traditional Chinese throughout the interface.

Do NOT use Simplified Chinese.

10. Equipment Selection

After selecting an exercise, the system should present the relevant equipment options where appropriate.

Examples:

- Bodyweight
- Barbell
- Dumbbell
- Kettlebell
- Cable
- Hex Bar / Trap Bar
- Machine
- etc.

Equipment options should be contextual rather than presenting an unnecessarily large universal list.

Some exercises have predefined equipment requirements.

For example:

Deadlift

- Barbell
- Trap Bar / Hex Bar
- etc.

Face Pull

- Cable

Some sports or activities do not need an equipment selection step at all.

The UI should intelligently skip irrelevant steps.

11. Weight Training Data Model

Every individual set must be stored as an independent record.

Example:

Flat Bench Press

Set 1:

60 kg × 10 reps

Set 2:

60 kg × 10 reps

Set 3:

70 kg × 8 reps

These are three individual set records.

Do NOT store only:

Bench Press — 60kg × 10 × 2 + 70kg × 8

The underlying data must preserve individual sets.

This is necessary for future analytics.

12. Dumbbell Convention

For dumbbell exercises, the entered weight represents the **total weight**, NOT the weight of one dumbbell.

For example:

Dumbbell Bench Press — 30 kg

means:

30 kg total

Do not interpret 30 kg as 30 kg per hand.

This convention should be clearly communicated in the UI where appropriate.

13. Bodyweight + Additional Weight

ULUX must support bodyweight exercises with additional external load.

Examples:

Pull-up

Bodyweight × 8

or

Bodyweight + 10 kg × 6

The data model should distinguish between:

- Bodyweight
- Additional external weight

This is important for exercises such as:

- Pull-ups
- Dips
- Push-ups
- Weighted squats
- Other bodyweight exercises

Do not simply force bodyweight exercises into a standard "weight" field.

14. Other Exercise Data Types

Not all activities use weight × reps.

The data model must support different measurement formats depending on the activity.

Examples:

Running

Distance + Duration

Cycling

Distance + Duration

Swimming

Distance + Duration

Tennis

Duration

Basketball

Duration

The architecture should therefore use an extensible activity-data model rather than assuming every exercise has:

weight + reps

However, do NOT add unnecessary metrics such as:

- Heart rate
- Calories
- Pace
- GPS
- Tempo
- RPE
- RIR

in the MVP unless needed for the basic activity type.

15. Notes

Workout records should support an optional free-text notes field.

Notes can be used for anything the user wants.

Examples:

"Felt unusually strong today."

"Used a different bench."

"Weight is total weight."

Do not create multiple unnecessary metadata fields in the MVP.

16. Custom Exercises

Every user can create their own exercises.

A custom exercise only needs:

- Exercise name
- Optional notes

The user does NOT need to define a complicated taxonomy when creating the exercise.

Once created, the custom exercise should appear in that user's exercise selection interface in future workouts.

Custom exercises are private to that user.

They should NOT automatically become global exercises.

17. Editing Custom Exercises

Users can:

- Edit custom exercise name
- Edit custom exercise notes
- Delete custom exercises if appropriate

Important:

If a custom exercise is renamed, its historical records should reflect the updated exercise name.

Historical workout records should remain valid.

Avoid destructive database behavior where editing a custom exercise breaks historical records.

Prefer a relational reference to the exercise entity so historical records naturally reflect the updated name.

18. Workout Editing

Users must be able to edit historical workout data.

They should be able to:

- Edit an individual set
- Edit weight
- Edit reps
- Edit duration/distance where applicable
- Edit notes
- Add sets
- Delete individual sets
- Edit workout details
- Delete an entire Workout Session

Deletion of an entire Workout Session should require an appropriate confirmation.

19. Repeat Last Workout

ULUX should learn from the user's previous training history.

When starting a workout, the user should be able to access:

Repeat Last Workout

For example, if the user's previous workout contained:

- Flat Bench Press
- Incline Bench Press
- Cable Fly
- Triceps Pushdown

the user can load that workout as a template for today's session.

The user must then be able to modify:

- Exercises
- Sets
- Weight
- Reps
- Notes
- etc.

The previous workout itself must NOT be modified.

This creates a powerful "progressive logging" workflow.

20. Recent / Frequently Used Exercises

The UI should remember the user's exercise history.

Frequently/recently used exercises should be easy to access.

For example, after selecting Weight Training:

Recent

Flat Bench Press

Squat

Deadlift

Pull-up

This should make logging significantly faster without changing the underlying taxonomy.

21. Home Screen

The Home screen should be extremely minimal.

Primary action:

Start Workout

Secondary useful actions may include:

Repeat Last Workout

and a small section for recently used exercises.

Do NOT turn the Home screen into a dense analytics dashboard.

The user should be able to start logging within seconds.

22. Calendar / History

ULUX should provide a calendar-based history view.

The calendar should clearly indicate days on which the user completed workouts.

Users should be able to:

- View monthly calendar
- See workout days
- Select a specific date
- View workouts on that date
- Open a specific Workout Session
- Inspect individual sets/details

The interface should also support a weekly view.

23. Workout Statistics

The system should calculate at minimum:

Weekly

- Number of workout days
- Number of workout sessions

Monthly

- Number of workout days
- Number of workout sessions

Be careful to distinguish:

Workout days

from:

Workout sessions

For example, two workouts on the same day count as:

- 1 workout day
 - 2 workout sessions
-

24. Progress / Analytics

ULUX should include a dedicated Progress / Analytics area.

This should NOT overwhelm the Home screen.

Potential analytics include:

Exercise progression

For example:

Flat Bench Press

Show historical progression of:

- Weight
- Reps
- Estimated strength / PR where appropriate

Personal Records

Track PRs where meaningful.

For strength exercises, useful PR types may include:

- Maximum weight
- Highest relevant load
- Potential estimated 1RM

Do not overcomplicate PR calculations in the first version.

Training volume

Where meaningful:

weight × reps

can be used to calculate training volume.

Training frequency

Examples:

- Workouts this week
- Workouts this month
- Training days
- Sessions per week

Muscle distribution

The system should be able to analyze training distribution across muscle groups.

However:

Muscle distribution should NOT be shown prominently on the initial Dashboard.

It should be accessible through advanced analytics.

Because exercises may belong to multiple muscle groups, avoid misleading calculations caused by double-counting. The implementation should use a sensible weighting or clearly defined methodology.

25. Data Visualization

Progress charts should be clean and easy to understand on mobile.

Examples:

- Exercise weight progression
- Volume progression
- Training frequency
- PR history

Avoid excessive charts.

The goal is:

useful insight, not a data science dashboard.

Charts should only be shown when sufficient data exists.

26. Excel Export

Users must be able to export their own complete workout history to an Excel `.xlsx` file.

The exported file should be useful outside ULUX, not simply a raw database dump.

A recommended workbook structure is:

Sheet 1 — Workout Summary

Columns such as:

- Date
- Workout Session
- Training Type
- Duration if available
- Notes

Sheet 2 — Exercise Records

Each set/activity record should be represented individually.

Possible columns:

- Date
- Workout Session
- Training Type
- Category
- Exercise
- Equipment
- Set Number
- Weight
- Additional Weight
- Reps
- Duration
- Distance
- Notes

Only relevant fields need to be populated for each activity.

Sheet 3 — Personal Records

Include meaningful PR information where available.

Sheet 4 — Statistics

Include useful aggregate statistics where appropriate.

The exact spreadsheet implementation can be adjusted for usability, but the export should preserve the underlying data without loss.

27. Mobile-First Design

Mobile is the primary platform.

The majority of logging is expected to happen:

inside a gym, on a smartphone, during a workout.

Therefore:

- Large touch targets
- Minimal typing
- Fast navigation
- Minimal scrolling
- Clear hierarchy
- Numeric input optimized for mobile
- Easy set creation
- Easy editing
- Easy repetition of previous workouts

The application should still work well on desktop, but desktop is secondary.

28. UI / Visual Direction

The visual language should be:

- Minimal
- Modern
- Clean
- Quiet
- Functional
- Premium but not luxurious
- Data-oriented without feeling technical

Avoid:

- Excessive gradients
- Gamification
- Giant fitness imagery
- Aggressive bodybuilding aesthetics
- Excessive cards
- Overloaded dashboards
- Unnecessary animations

ULUX should feel like a thoughtful piece of software rather than a generic fitness app template.

29. Traditional Chinese

The primary interface language is **Traditional Chinese (繁體中文)**.

All user-facing interface copy should use Traditional Chinese.

Exercise names should generally use:

Traditional Chinese — English

For example:

深蹲 — Squat
硬舉 — Deadlift
平胸臥推 — Flat Bench Press
上胸臥推 — Incline Bench Press
引體向上 — Pull-up
臉拉 — Face Pull

Do not use Simplified Chinese.

English may be used alongside Chinese where useful.

30. Exercise Taxonomy Architecture

The exercise database should be designed as structured data rather than hardcoded UI strings.

At minimum, an exercise should be capable of having:

- ID
- Traditional Chinese name
- English name
- Exercise type
- One or more muscle-group relationships
- Equipment relationships
- Measurement/data format
- Whether it is a system exercise or user-created exercise
- Optional notes

Muscle groups and exercises should support many-to-many relationships.

For example:

Face Pull → Shoulders
Face Pull → Back

This makes the system extensible.

31. System Exercises vs Custom Exercises

There are two categories of exercises.

System exercises

Built into ULUX.

These should cover common strength exercises and common cardio/sports activities.

User exercises

Created privately by individual users.

User-created exercises should only appear for their creator.

Do not modify the global exercise database when a user creates or edits a custom exercise.

32. Architecture Requirements

Use a clean architecture that separates:

- Authentication
- Users
- Exercises
- Muscle groups
- Equipment
- Workout Sessions
- Workout records / sets
- Analytics
- Export

The data model should be normalized enough to prevent duplication and preserve historical integrity.

Do not hardcode exercise relationships directly into frontend components.

The exercise taxonomy should be data-driven.

33. Future Extensibility

The architecture should leave room for future features such as:

- AI training analysis
- AI-generated insights
- More sophisticated progression analysis
- Wearable integrations
- Apple Health integration
- Garmin / Strava integration
- More detailed exercise metrics

However:

DO NOT IMPLEMENT THESE FEATURES IN THE MVP.

The current product should focus on excellent core logging and historical data.

34. MVP Scope

The MVP MUST include:

Authentication

- Account creation
- ID
- 4-digit PIN
- Email
- Nickname
- Login
- Logout
- PIN recovery through email
- Immutable ID

Workout logging

- Workout Session
- Training type
- Category
- Exercise
- Equipment where applicable
- Individual sets
- Weight
- Reps
- Bodyweight + additional weight
- Duration/distance where applicable
- Notes

Exercise system

- Built-in exercises
- Traditional Chinese + English names
- Multiple muscle-group relationships
- User-created exercises
- Edit custom exercises

Workout management

- Create
- Edit
- Delete individual records
- Delete Workout Session
- Repeat Last Workout

History

- Calendar
- Weekly view
- Date details
- Workout Session details

Statistics

- Weekly workout days
- Monthly workout days
- Basic training frequency
- Basic volume
- PRs
- Exercise progression

Analytics

- Progress charts
- Advanced muscle distribution

Account

- Settings
- Change PIN
- Nickname
- Manage custom exercises
- Excel export
- Logout

UX

- Mobile-first
- Responsive desktop support
- Fast workout entry

35. Explicitly Out of Scope for MVP

Do NOT implement:

- Social networking
- Public profiles
- Followers
- Likes
- Comments
- Messaging
- Leaderboards
- Gamification
- Nutrition tracking
- Meal tracking

- Calories
- Supplements
- Heart-rate integration
- Wearable integrations
- Apple Health integration
- Garmin integration
- Strava integration
- AI coaching
- AI-generated workout plans
- Push notifications
- Complex subscription/payment systems

Unless specifically requested later, these should not be added.

36. Product Decision Rule for the Coding Agent

When a requirement is explicitly defined in this document:

Follow it exactly.

When a detail is intentionally unspecified:

Make the simplest reasonable implementation that is consistent with ULUX's product philosophy.

Do NOT introduce major new product features simply because they are technically possible.

Do NOT over-engineer the MVP.

Do NOT add unnecessary configuration options.

Do NOT create unnecessary screens.

Prefer:

simple UI + robust underlying data model

over:

complicated UI + shallow data structure.

37. Important Data Integrity Rules

Historical workout data is valuable.

Therefore:

- Editing an exercise should not destroy historical records.
 - Deleting a workout should not accidentally delete unrelated workouts.
 - Deleting a custom exercise should be handled carefully if historical records reference it.
 - Every set must remain independently identifiable.
 - Multiple sessions on the same date must remain distinguishable.
 - User data must remain isolated between accounts.
 - Custom exercises must remain private to their creator.
 - Analytics must be derived from structured underlying records rather than duplicated manually maintained values.
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38. Security

Although ULUX intentionally has minimal authentication, user data must still be protected.

Implement:

- Secure password/PIN hashing
- Secure session management
- Proper authorization checks
- User-level data isolation
- Protection against unauthorized access to another user's workout data
- Secure email-based recovery
- Server-side validation

Never rely exclusively on frontend checks for authorization.

39. Suggested Primary Navigation

Keep the navigation simple.

A possible structure:

Home

Start Workout
Repeat Last Workout
Recent Exercises

History

Calendar
Weekly / Monthly history
Workout Sessions

Progress

- Exercise Progress
- PRs
- Training Volume
- Advanced Analytics
- Muscle Distribution

Settings

- Nickname
- Change PIN
- Custom Exercises
- Export Excel
- Logout

The exact navigation structure can be adjusted if a better mobile UX is found, but the information architecture should remain simple.

40. Primary User Flow

The ideal first-time returning user flow should feel like:

```
graph TD; A[Open ULUX] --> B[Login with ID + PIN]; B --> C[Home]; C --> D[Start Workout]; D --> E[Choose Training Type]; E --> F[Choose Category]; F --> G[Choose Exercise]; G --> H[Choose Equipment if applicable]; H --> I[Enter first set]; I --> J[Add additional sets]; J --> K[Add notes if desired]; K --> L[Save]; L --> M[Workout completed];
```

For returning users:

Open ULUX
↓
Login
↓
Repeat Last Workout
↓
Modify today's sets
↓
Save

This second flow should be extremely fast.

41. Success Criteria

The MVP should be considered successful if a regular strength-training user can:

1. Create an account in a few minutes.
2. Log into ULUX quickly using ID + PIN.
3. Start a workout without navigating through unnecessary screens.
4. Record individual sets accurately.
5. Repeat a previous workout.
6. Modify today's workout without modifying historical data.
7. Find any previous workout through the calendar.
8. Understand their basic training frequency.
9. See how an exercise has progressed over time.
10. Identify meaningful PRs.
11. Export their complete training history to Excel.
12. Perform all major workout logging tasks comfortably on a smartphone.

The most important metric is not the number of features.

It is:

How quickly and effortlessly can a user record a real workout while standing in the gym?

42. Final Product Principle

ULUX should feel like:

A notebook that remembers everything.

The user should never feel like they are entering data into a complicated database.

They should feel like they are simply recording their workout.

The complexity should exist underneath the interface so that ULUX can gradually transform those simple records into meaningful personal training intelligence.